Answerability-First RAG for Mixed Text and Tables

Pycon 2026 – Maria Vallarelli

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Use Case - Challenge

Why **answerability-first** changes the way we build **RAG systems**

1. WHEN YOU BUILD A RAG...

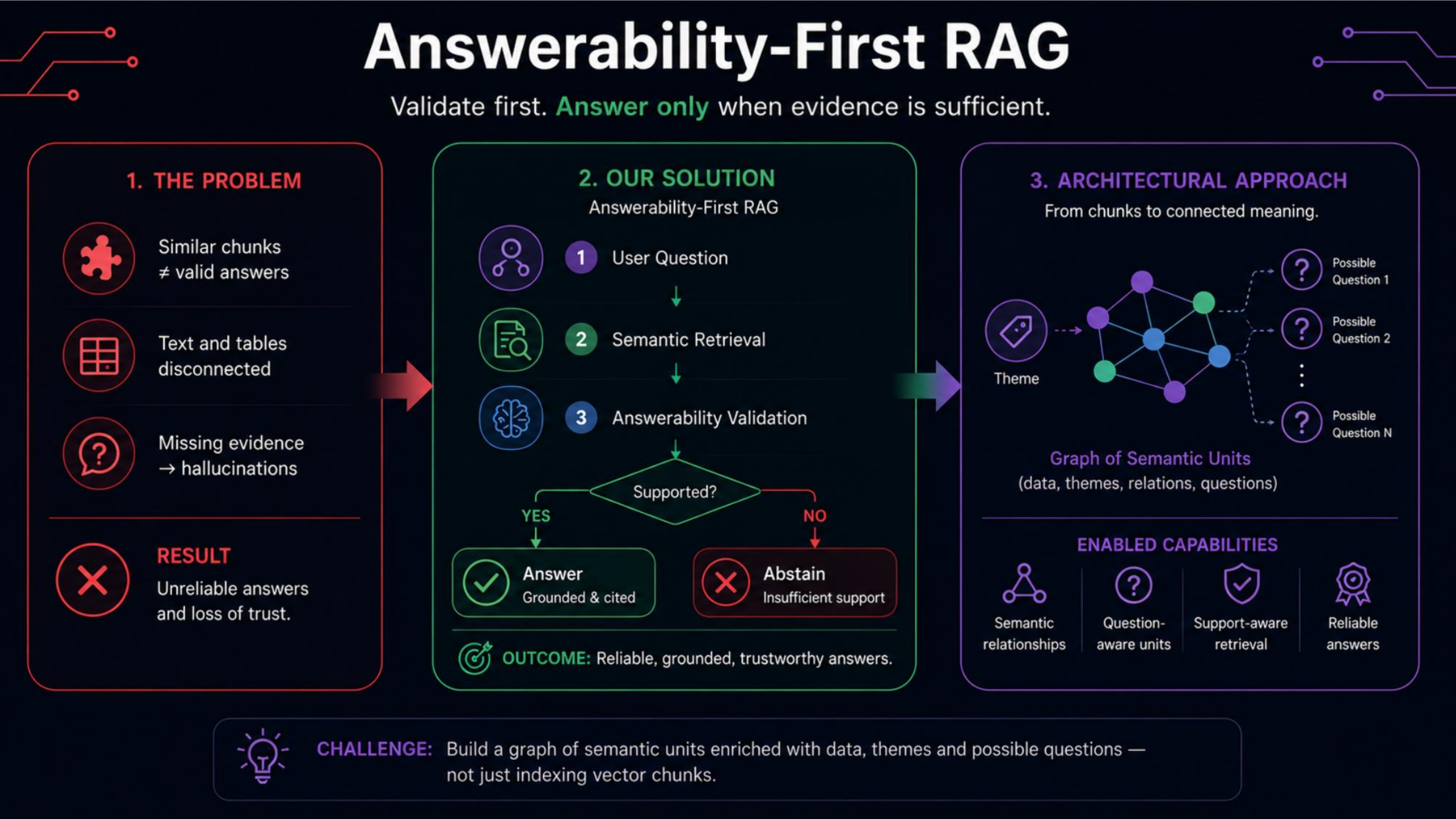
More data, more context, more embedding... fine!

2. REALITY CHECK

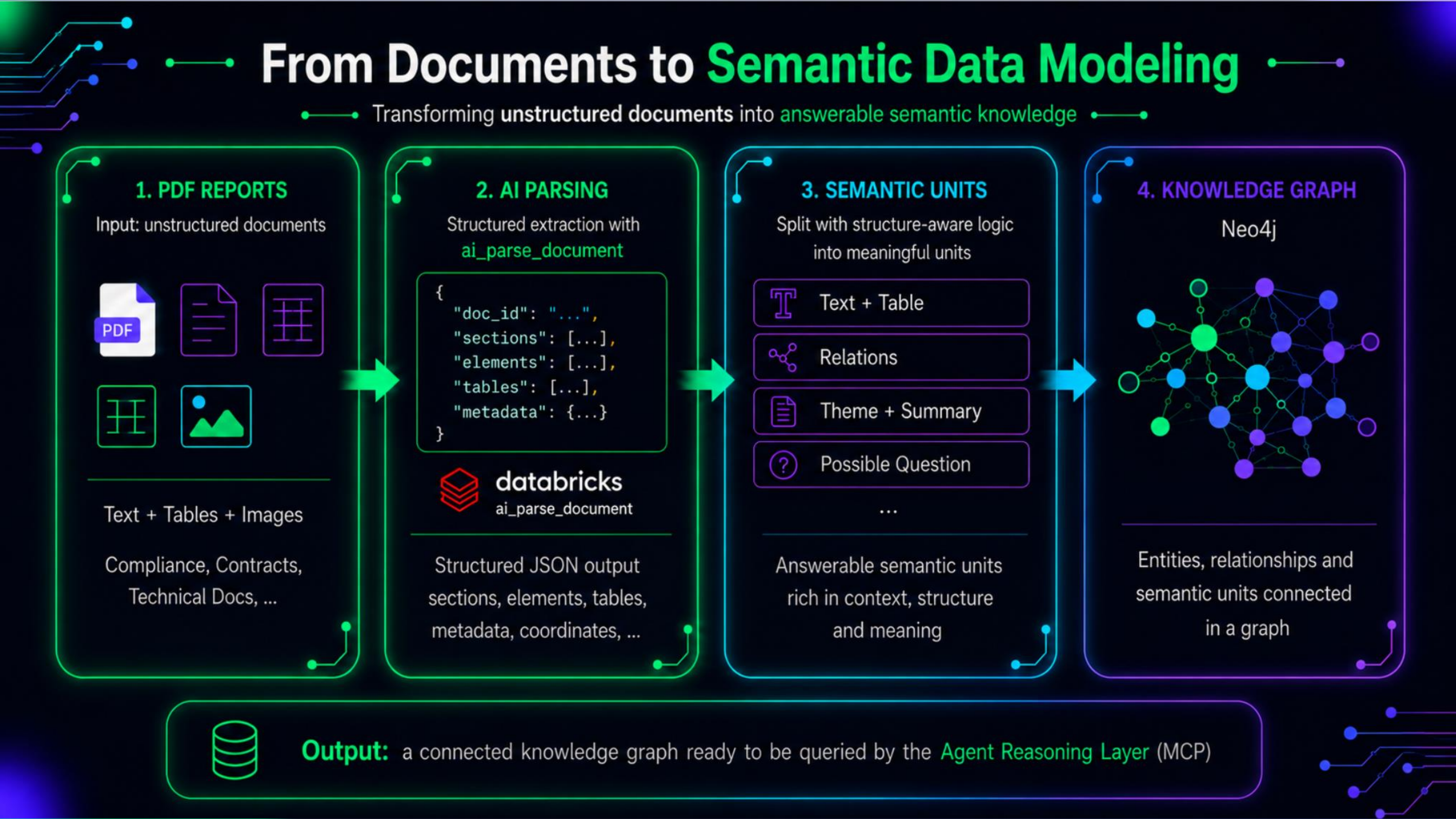
Hallucinations, unverifiable answers, zero confidence.

3. ANSWERABILITY-FIRST TO THE RESCUE

Answers you can verify.
Sources you can trust.
Impact you can measure.



Answerability-First RAG
Validate first. Answer only when evidence is sufficient.
1. THE PROBLEM
Similar chunks ≠ valid answers
Text and tables disconnected
Missing evidence → hallucinations
RESULT
Unreliable answers and loss of trust.
2. OUR SOLUTION
Answerability-First RAG
1 User Question
2 Semantic Retrieval
3 Answerability Validation
Supported?
YES
NO
Answer
Grounded & cited
Abstain
Insufficient support
OUTCOME: Reliable, grounded, trustworthy answers.
3. ARCHITECTURAL APPROACH
From chunks to connected meaning.
Theme
Possible Question 1
Possible Question 2
Possible Question N
Graph of Semantic Units
(data, themes, relations, questions)
ENABLED CAPABILITIES
Semantic relationships
Question-aware units
Support-aware retrieval
Reliable answers
CHALLENGE: Build a graph of semantic units enriched with data, themes and possible questions — not just indexing vector chunks.


From Documents to Semantic Data Modeling
Transforming unstructured documents into answerable semantic knowledge
1. PDF REPORTS
Input: unstructured documents
PDF
Text + Tables + Images
Compliance, Contracts,
Technical Docs, ...
2. AI PARSING
Structured extraction with
ai_parse_document
{
"doc_id": "...",
"sections": [...],
"elements": [...],
"tables": [...],
"metadata": {...}
}
databricks
ai_parse_document
Structured JSON output
sections, elements, tables,
metadata, coordinates, ...
3. SEMANTIC UNITS
Split with structure-aware logic
into meaningful units
Text + Table
Relations
Theme + Summary
Possible Question
...
Answerable semantic units
rich in context, structure
and meaning
4. KNOWLEDGE GRAPH
Neo4j
Entities, relationships and
semantic units connected
in a graph
Output: a connected knowledge graph ready to be queried by the Agent Reasoning Layer (MCP)

2 ai_parse_document – Example Output

databricks

From unstructured PDF to structured, AI-parsed elements

INPUT: UNSTRUCTURED PDF

Quarterly Financial Results
Q1 2024

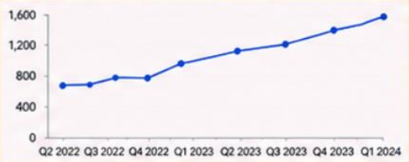
1. Executive Summary

The company delivered strong performance in Q1 2024, with significant growth across all key metrics.

Key Financial Highlights

Metric	Q1 2023	Q1 2024	YoY Change
Revenue ($M)	1,250	1,580	+26%
Gross Profit ($M)	620	810	+31%
Operating Income ($M)	210	315	+50%
Net Income ($M)	160	250	+56%

Figure 1. Revenue growth over the last 8 quarters.



2. Business Update

Demand remains high across all regions. We continue to invest in product and operational excellence.

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DETECTED ELEMENTS (examples)

- title
- section_header
- text
- table
- figure
- caption
- section_header
- text
- page_footer

OUTPUT: STRUCTURED JSON (simplified example)


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{
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    "pages": [
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      { "id": 1, "image_uri": "pages/page_1.png" },
      { "id": 2, "image_uri": "pages/page_2.png" }
    ],
    "elements": {
      {
      { "id": 0, "type": "content"; "Quarterly Financial Results", "confidence": 0.99,
        "bbox": [{ "coord": [72, 72, 540, 110], "page_id": 0 }], "description": "Document title"}
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        with significant growth across all key metrics.", "confidence": 0.95,
        "bbox": [{ "coord": [72, 150, 540, 190], "page_id": 0 }], "description": "Executive summary text",
      },
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        "confidence": 0.98, "bbox": [{ "coord": [72, 210, 544, 350], "page_id": 0 }],
        "description": "Financial metrics comparsion table" }
      },
      { "id": 3, "type": "caption", "content": "Figure 1. Revenue growth over the last 8 quarters.",
        "confidence": 0.97, "bbox": [{ "coord": [72, 360, 540, 385], "page_id": 0 }],
        "description": "Caption for the line chart below" }
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        "bbox": [{ "coord": [72, 390, 544, 520], "page_id": 0 }],
        "description": "Line chart with upward trend across 8 quarters" }
      }.. more elements
    }
  },
  "error_status": [],
  "metadata": {
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      "file_name": "Q1_2024.pdf",
      "file_size": 2456789,
      "file_modification_time": "2024-05-15T10:30:00Z"
    }
}
```

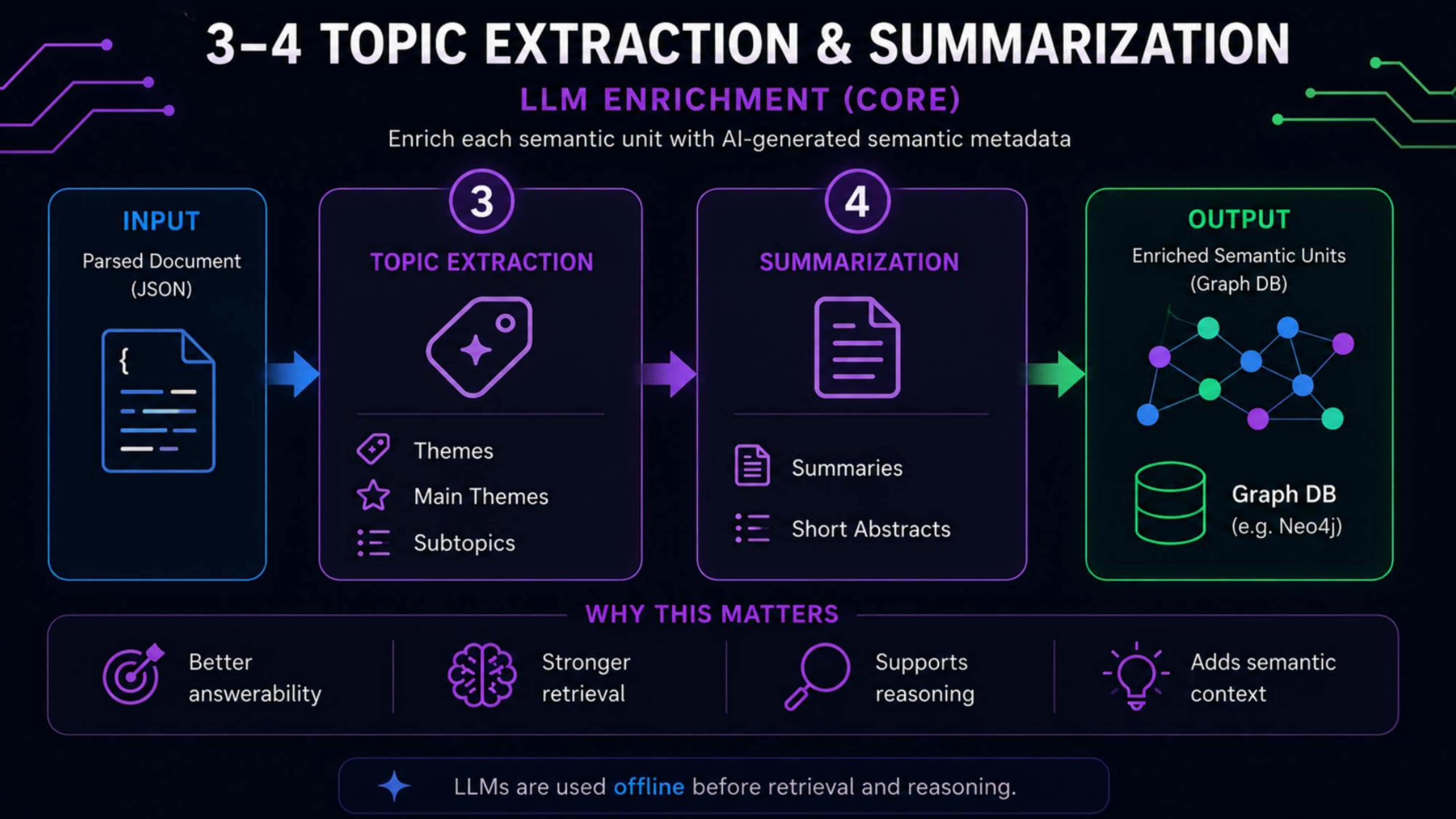

WHAT THIS GIVES US

Precise Location
Each element has bounding box (bbox) and page_id

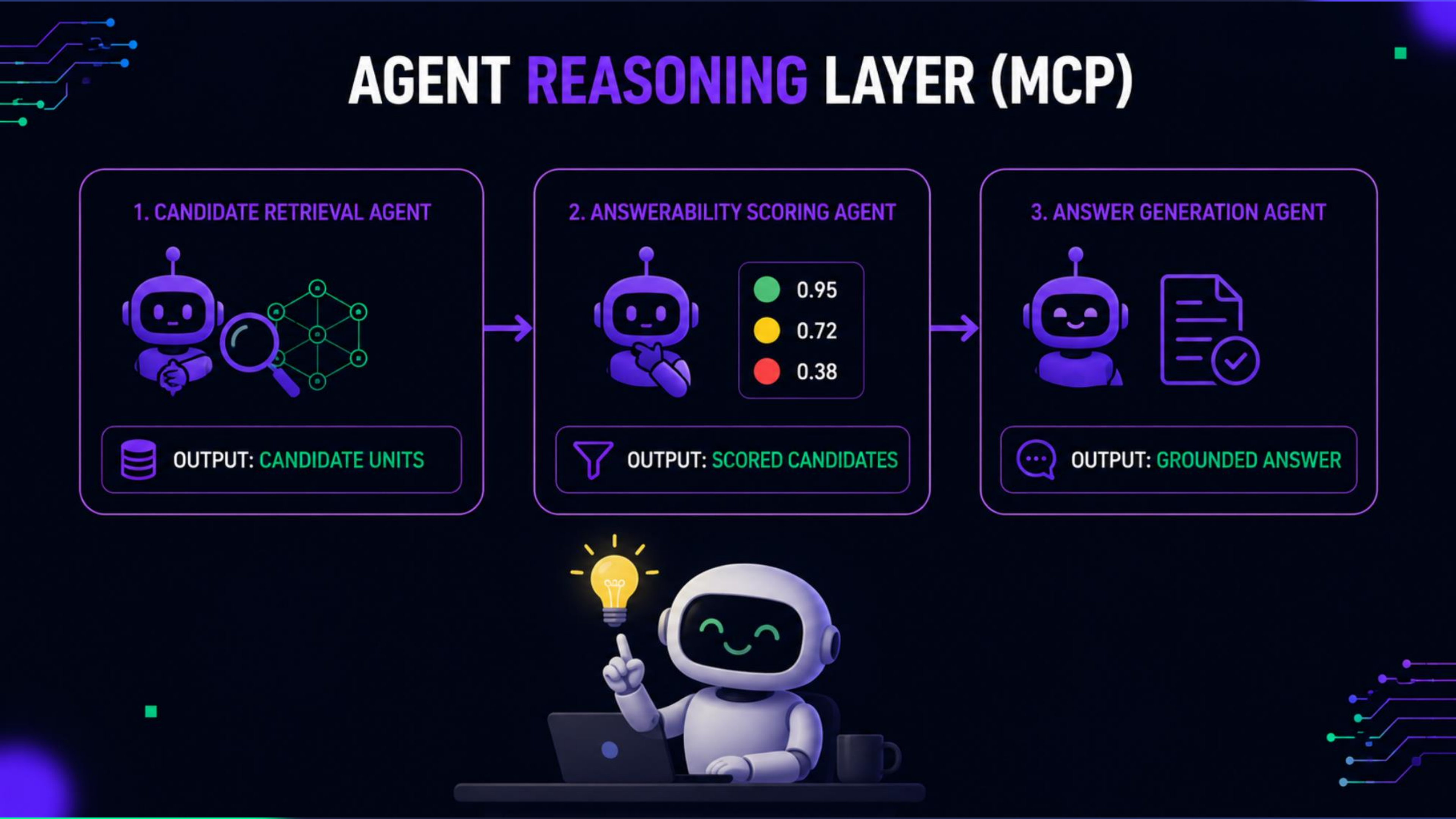
Rich Semantics
Element type, content, confidence and descriptions

Ready for Graph
Ideal to build the knowledge graph in GraphDB

Traceable & Explainable
Every answer can be linked back to the original source

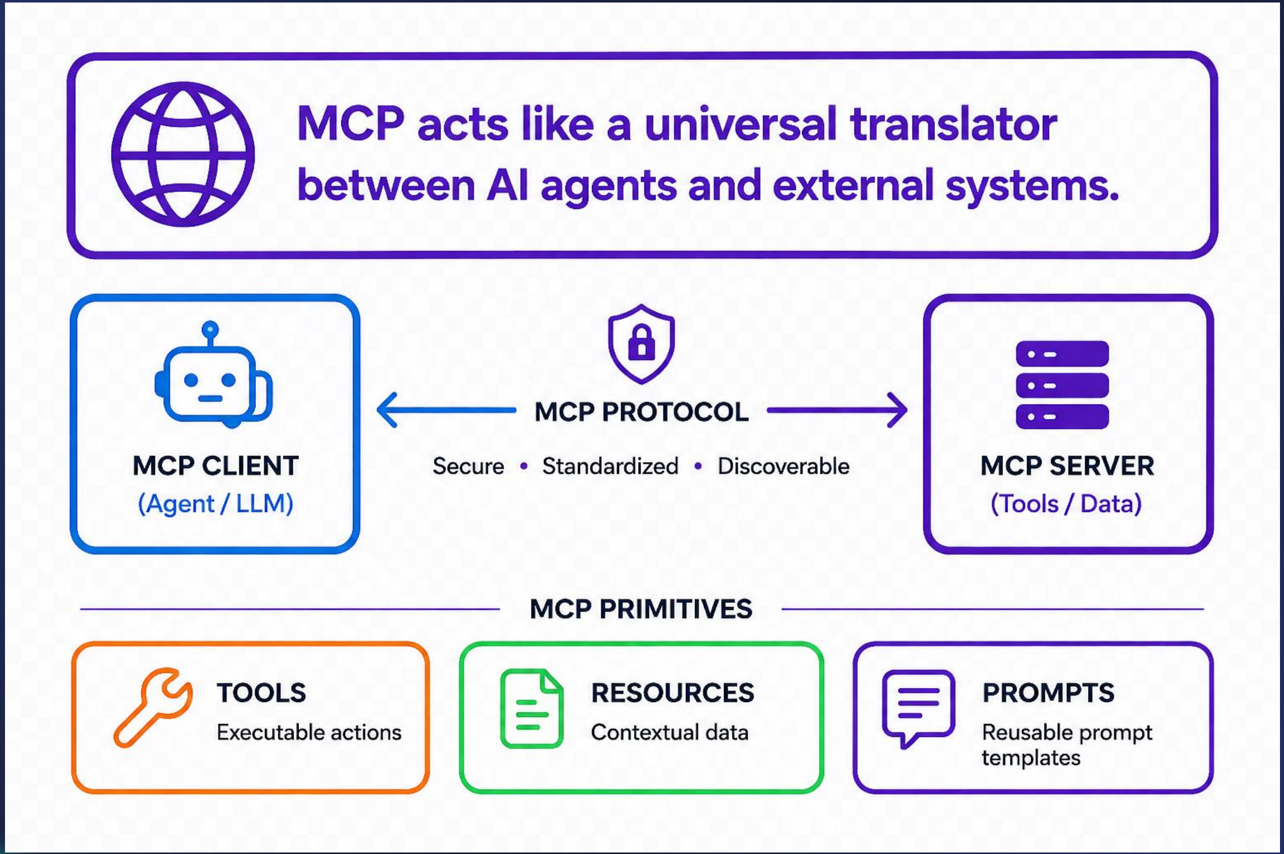


3–4 TOPIC EXTRACTION & SUMMARIZATION
LLM ENRICHMENT (CORE)
Enrich each semantic unit with AI-generated semantic metadata
INPUT
Parsed Document (JSON)
3
TOPIC EXTRACTION
Themes
Main Themes
Subtopics
4
SUMMARIZATION
Summaries
Short Abstracts
OUTPUT
Enriched Semantic Units (Graph DB)
Graph DB
(e.g. Neo4j)
WHY THIS MATTERS
Better answerability
Stronger retrieval
Supports reasoning
Adds semantic context
LLMs are used offline before retrieval and reasoning.



AGENT REASONING LAYER (MCP)
1. CANDIDATE RETRIEVAL AGENT
OUTPUT: CANDIDATE UNITS
2. ANSWERABILITY SCORING AGENT
0.95
0.72
0.38
OUTPUT: SCORED CANDIDATES
3. ANSWER GENERATION AGENT
OUTPUT: GROUNDED ANSWER

Understanding MCP



QA Pipeline Architecture

USER LAYER: Marimo UI

Users ask questions and view answers

↓

AGENT LAYER: PydanticAI MCP Client

Creates the QA Agent

↓

MCP SERVER LAYER: FastMCP

1.Semantic Retrieval 2. Answerability Validation
3.Grounded Answer Generation

↓

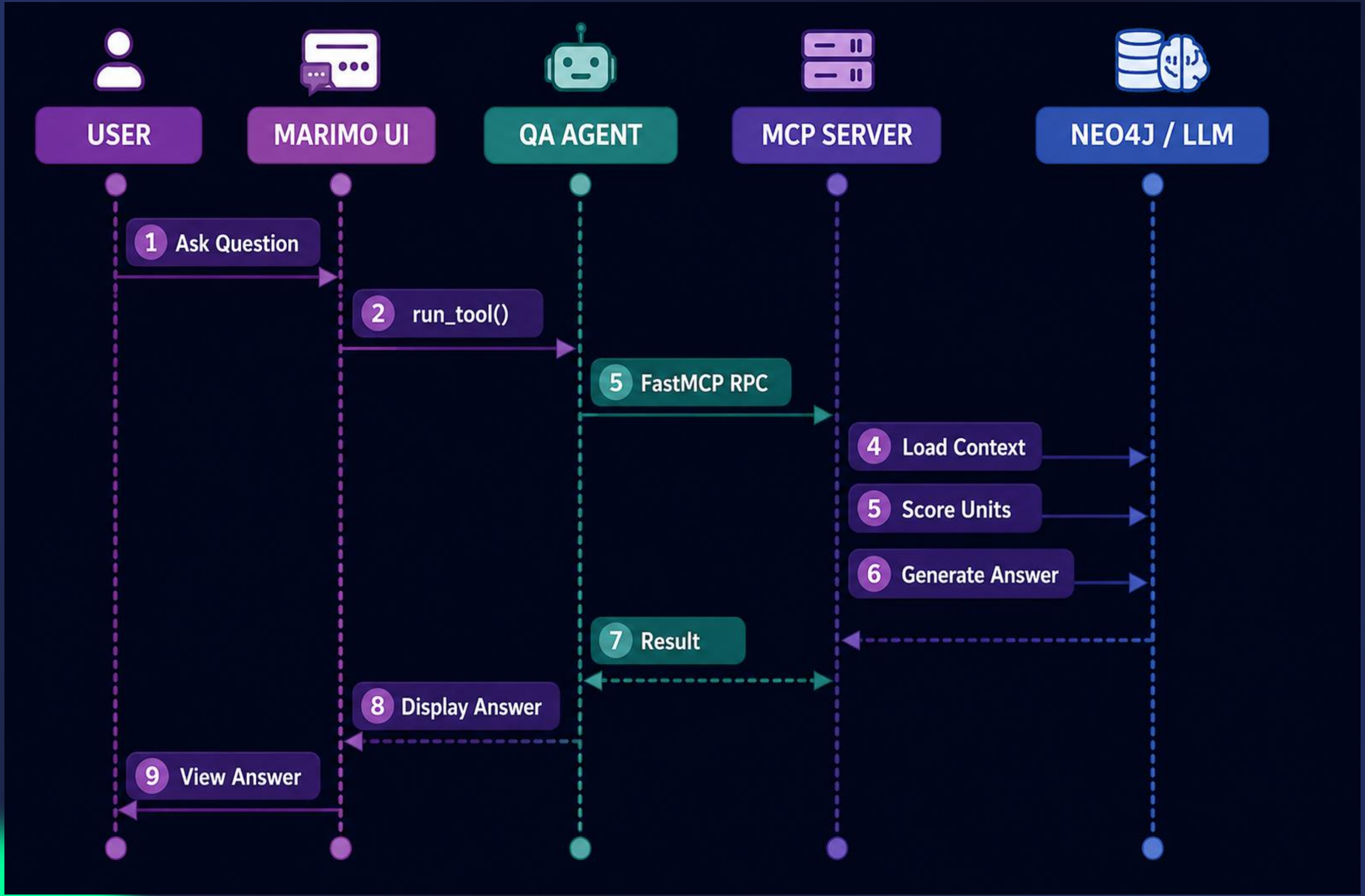
NEO4J CONTEXT DB

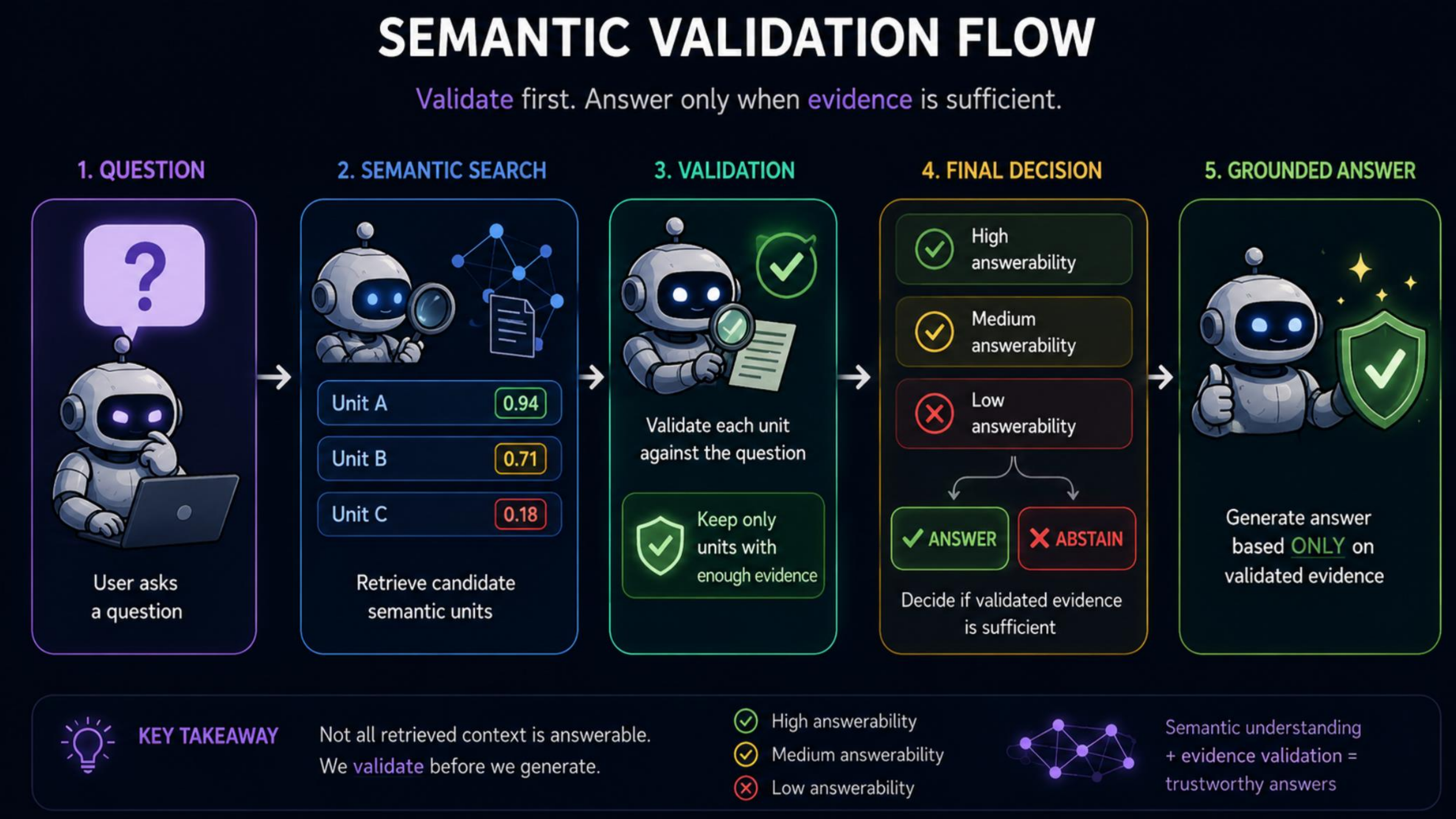
Stores and retrieves entities and relationships

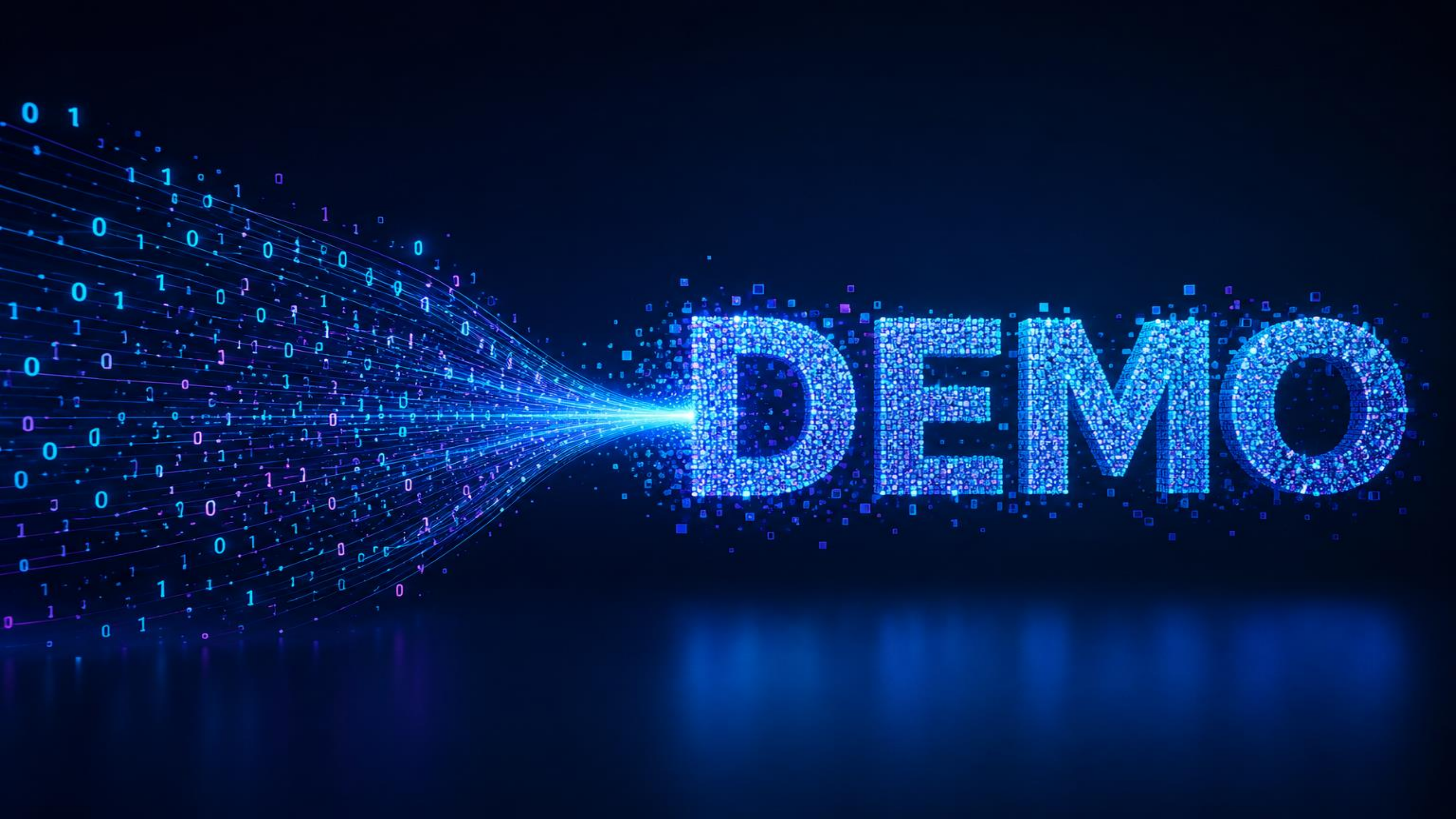
GROUND TRUTH VALIDATION:

Expected units to validate answerability

Sequence Diagram (RPC flow)




SEMANTIC VALIDATION FLOW
Validate first. Answer only when evidence is sufficient.
1. QUESTION
?
User asks a question
2. SEMANTIC SEARCH
Unit A 0.94
Unit B 0.71
Unit C 0.18
Retrieve candidate semantic units
3. VALIDATION
Validate each unit against the question
Keep only units with enough evidence
4. FINAL DECISION
High answerability
Medium answerability
Low answerability
ANSWER
ABSTAIN
Decide if validated evidence is sufficient
5. GROUNDED ANSWER
Generate answer based ONLY on validated evidence
KEY TAKEAWAY
Not all retrieved context is answerable.
We validate before we generate.
High answerability
Medium answerability
Low answerability
Semantic understanding + evidence validation = trustworthy answers


DEMO

References

[Answerability in Retrieval-Augmented Open-Domain Question Answering]

[Groundedness in Retrieval-augmented Long-form Generation: An Empirical Study]

[SURE-RAG: Sufficiency and Uncertainty-Aware Evidence Verification]

[Assessing the Answerability of Queries in Retrieval-Augmented Code Generation]

[Databricks - Ai_parse_document]

[Tools vs Resources vs Prompts in MCP]

[Pydantic Docs]

💻 Demo & project repository → https://github.com/mariavallarelli/answerability-first-rag

Thanks!

SCAN ME